

Assignment 2 Tim Hortons

(a) Algorithm Design

Items necessary: K highway length, L max distance, H list of off ramps (lets call individual offramps as $H[i]$)

We need to store the values for the chosen ramps lets call it S

Need a value to indicate rightmost point that is covered by a location lets call it x

While loop will follow if coverage is less than $K - L$

Needs to consider all off ramps within L such that $H[i] \leq \text{coverage} + L$ and $H[i]$ needs to be further than current coverage

From this pick furthest location possible

Add to S

Update coverage to = best ramp

*came back to this because I didnt notice this was necessary until I saw the sample output of the example

Check if $H[n-1]$ where n is the length of H is in S and if not add to S

Pseudocode(K, L, H):

$S = []$

Covered = 0

$n = \text{len}(H)$

While Covered < ($K - L$):

$\text{Max} = \text{null}$

 While $i < n$ && $H[i] \leq \text{Covered} + L$:

 If $H[i] > \text{Covered}$:

$\text{Max} = H[i]$

$i = i + 1$

```
S.append(Max)
Covered = Max
```

```
If H[n-1] != S[len(S) - 1]
S.append(H[n-1])
```

(b) Proof of Optimality

Suppose the greedy solution S is not optimal

Suppose there is an optimal solution S^* of size n

S^* and S have differing locations at some point in H

Let $H[g]$ be the off ramp location chosen by greedy algo to cover a point x , the optimal solution has some different $H[s]$ to cover that point of the highway

Because of greedy algorithms $H[g]$ is the rightmost location that can still cover x , $H[g] > H[s]$

We can replace $H[s]$ in S^* with $H[g]$, and thus the coverage of x and all points up to $H[g]$ are now included and because $H[g] > H[s]$ there is no coverage lost

S^* and S now look closer

Repeating this replacement at all locations in H of S^* with H of S causes the solution S^* to be eventually equal to that of S but still has at most a size of n (now $|S| \leq n$)

Thus the optimal solution is the greedy solution S